K.R. MANGALAM UNIVERSITY
THE COMPLETE WORLD OF EDUCATION

Fundamentals of Operating System Lab (ENCA252)

SOET

Assignments-04

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BCA (AI & DS) (Research)

Task 6: Performance Comparison

The following table calculates the performance for the specific input: **Initial Head = 53**, **Disk Size = 200**, and **Queue = [98, 183, 37, 122, 14, 124, 65, 67]**.

Algorithm	Total Seek Time	Efficiency
FCFS	640	**Low**: Services in order of arrival
SSTF	236	**High**: Picks closest request next
SCAN	331	**Moderate**: Moves like an elevator
C-SCAN	382	**Moderate**: Moves in one circular direction

Task 7: Result Analysis

Comparison Analysis

For the given request sequence, the **SSTF (Shortest Seek Time First)** algorithm is the most efficient, requiring only **236 cylinders** of movement. This is a significant reduction compared to **FCFS**, which requires **640 cylinders** due to its lack of optimization. **SCAN** and **C-SCAN** provide balanced performance, with SCAN (331) being slightly more efficient than C-SCAN (382) in this specific case because it reverses direction rather than jumping back to the start.

Pros and Cons

- **FCFS (First Come First Serve)**:
  - **Pros**: Fair and simple to implement.
  - **Cons**: Highest seek time (convoy effect).
- **SSTF (Shortest Seek Time First)**:
  - **Pros**: Minimizes seek time by servicing the nearest request.
  - **Cons**: Can cause **starvation** for requests far from the current head position.
- **SCAN (Elevator Algorithm)**:
  - **Pros**: No starvation and good throughput.
  - **Cons**: Extreme edges may wait longer for the head to return.
- **C-SCAN (Circular SCAN)**:

- **Pros**: Provides a more **uniform waiting time** than SCAN.
- **Cons**: Higher seek time due to the non-servicing jump from end to start.

Conclusion

Based on the seek times calculated, **SSTF** is the best choice for pure efficiency. However, in real-world operating systems where fairness is required to prevent processes from waiting indefinitely, **SCAN or C-SCAN** is often preferred. These algorithms offer a more predictable response time across all disk cylinders while maintaining much better performance than the simple FCFS method.